Set 31: Solutions

Set 31: Exercises

1. A student recorded the solubility of sugar at different temperatures.

Temperature °C	Solubility (g/100 g water)
0	179
20	204
40	238
60	287
80	362
100	487

(a) Draw a solubility curve for sugar with temperature on the horizontal axis.
(b) Describe the pattern shown by the graph.
(c) Determine the solubility of sugar at:
   (i) 30 °C
   (ii) 70 °C
(d) Use the information in the student's table to describe an unsaturated, saturated and a super saturated solution of sugar at 20 °C?
(e) Classify the following sugar solutions as unsaturated, saturated or super-saturated.
   (i) 200 g of sugar dissolved in 100 g of water at 40 °C
   (ii) 200 g of sugar dissolved in 50 g of water at 60 °C
   (iii) 50 g of sugar dissolved in 20 g of water at 100 °C

2. A 2.000 kg (approximately 2.0 L) sample of recycled water on a property was examined for total dissolved solids, TDS. The purpose was to see if after treatment its TDS concentration was at an acceptable level for use in irrigation. The sample was evaporated to dryness and the total dissolved solids remaining weighed 3.45 g.
(a) Given that water up to 2,500 ppm can be used for irrigation determine the acceptability of the wastewater sample. Parts per million can be calculated as follows:

$$\text{ppm} = \frac{\text{mg of solute}}{\text{kg of solution}}$$

(b) Use the information right to classify the salinity of the sample:

Fresh water	Less than 1,000 ppm
Slightly saline water	From 1,000 ppm to 3,000 ppm
Moderatly saline water	From 3,000 ppm to 10,000 ppm
Highly saline water	From 10,000 ppm to 35,000 ppm

3. Seawater contains about 35,000 ppm of salt. How many grams of salt would be obtained if 1.000 kg (approximately 1 L) of seawater were evaporated to dryness?

4. Describe the chemical tests that would allow you to distinguish between solid samples of the following barium salts:
$BaCO_3$, $Ba(NO_3)_2$, $BaC\ell_2$ and $BaSO_4$.

5. Use the solubility table (Appendix 1) to identify and describe any precipitate that forms when the following solutions are mixed.
(a) $NaC\ell$ and $AgNO_3$
(b) $Pb(NO_3)_2$ and KI
(c) K_2SO_4 and $Ba(OH)_2$
(d) $CuSO_4$ and NaOH
(e) $(NH_4)_3PO_4$ and $FeC\ell_2$

6. Proceed to Set 27: Ionic equations. Read the explanations and examples. Write ionic equations for any precipitation reactions occurring in question 5.

7. Classify the following as strong, weak or non-electrolytes: tap water, sea water, sugar solution, copper sulfate solution, hydrochloric acid solution.

Notes